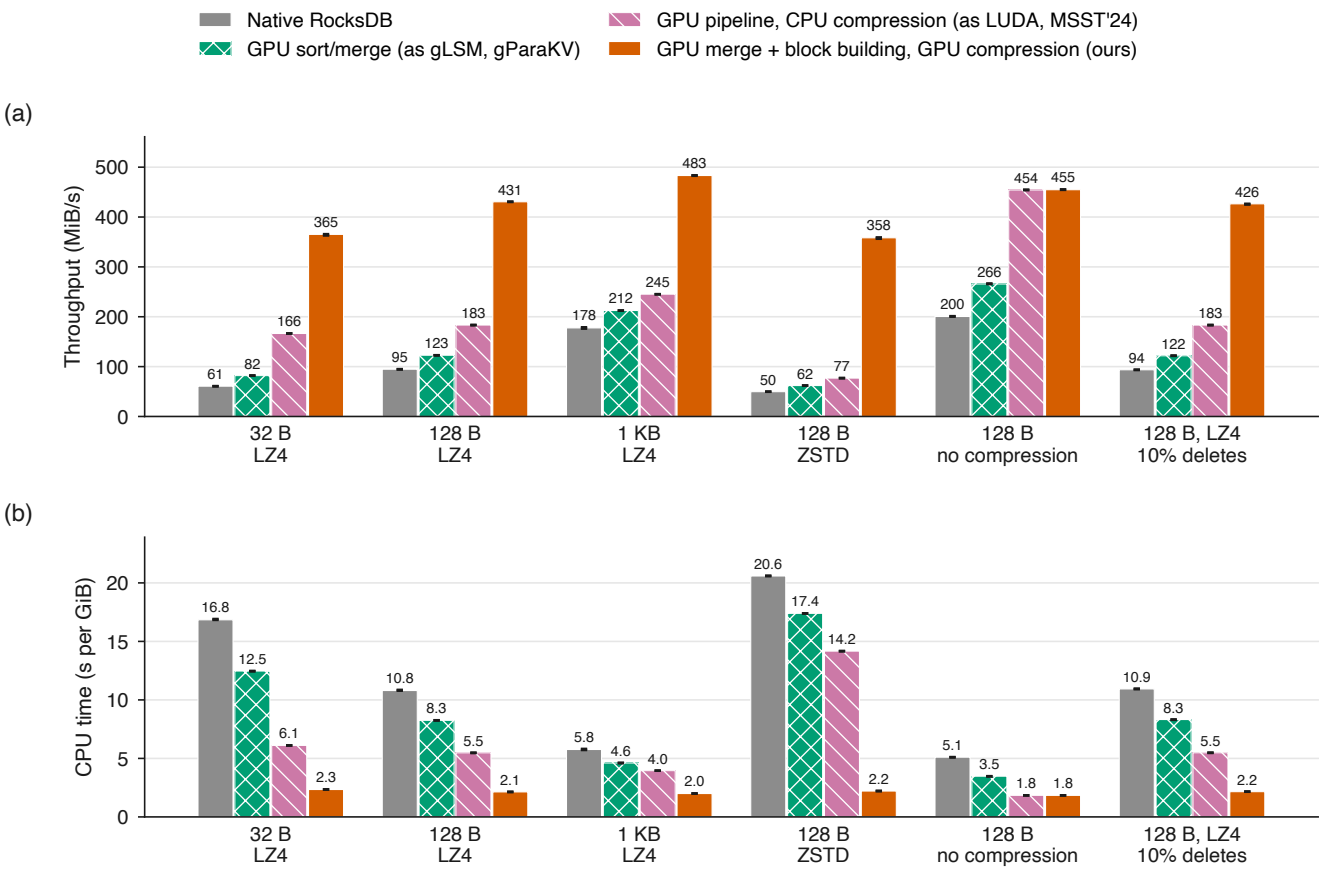
Native RocksDB
GPU sort/merge (as gLSM, gParaKV)
GPU pipeline, CPU compression (as LUDA, MSST'24)
GPU merge + block building, GPU compression (ours)
(a)
Throughput (MiB/s)
0
100
200
300
400
500
61
82
166
365
95
123
183
431
178
212
245
483
50
62
77
358
200
266
454
455
94
122
183
426
32 B LZ4
128 B LZ4
1 KB LZ4
128 B ZSTD
128 B no compression
128 B, LZ4 10% deletes
(b)
CPU time (s per GiB)
0
5
10
15
20
16.8
12.5
6.1
2.3
10.8
8.3
5.5
2.1
5.8
4.6
4.0
2.0
20.6
17.4
14.2
2.2
5.1
3.5
1.8
1.8
10.9
8.3
5.5
2.2
32 B LZ4
128 B LZ4
1 KB LZ4
128 B ZSTD
128 B no compression
128 B, LZ4 10% deletes